

Abhishek Gupta

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Professional Summary

Machine Learning and Risk Analytics professional with expertise in **Large Language Models, Statistical Modeling, and Model Validation**. Experienced in developing and evaluating predictive models using **Python, Deep Learning, and Econometric Techniques**. Skilled in **model interpretability, risk analytics, and AI governance** to ensure reliable, explainable, and high-performance machine learning systems. Proven ability to translate complex analytical results into business insights and support data-driven decision making.

Core Skills

Machine Learning	Supervised/Unsupervised Learning, Deep Learning, NLP, LLMs
Risk Analytics	Model Validation, Statistical Modeling, Volatility Modeling, Econometrics
GenAI / NLP	Retrieval-Augmented Generation (RAG), Text Mining, Transformers
Model Evaluation	Cross Validation, Bias Analysis, Model Explainability, Feature Engineering
Programming	Python, Java, SQL, PySpark
Big Data Tools	Hadoop (HDFS), MapReduce, Distributed Data Processing
Explainable AI	SHAP, Grad-CAM, Model Interpretability

Projects

Medical Research Summarization using Generative AI (RAG) Sept–Oct 2025

- Developed a **Retrieval-Augmented Generation pipeline** integrating BM25 and MiniLM embeddings for knowledge retrieval.
- Implemented **DistilBART-based abstractive summarization** for generating concise insights from medical research literature.
- Evaluated model outputs using semantic similarity metrics and improved retrieval relevance through multi-stage ranking.

Electricity Demand Forecasting using Hybrid Deep Learning Oct 2025

- Built a forecasting pipeline combining **1D CNN and LSTM architectures** for temporal energy demand prediction.
- Engineered time-series features and optimized hyperparameters to enhance predictive performance.
- Achieved **significant accuracy improvement over baseline ARIMA models**.

Vision Transformer based Plant Disease Detection Jun–Jul 2025

- Implemented **Vision Transformer (ViT) and Swin Transformer models** for multi-class plant disease classification.
- Applied **Grad-CAM explainability** to validate model decision regions and improve model interpretability.
- Processed large-scale agricultural image datasets and achieved strong F1 performance across multiple disease classes.

Financial Risk Modeling – Cryptocurrency Volatility Analysis Mar–Apr 2025

- Conducted econometric modeling using **GARCH(1,1) and EGARCH** to study asymmetric volatility behavior.
- Modeled heavy-tailed risk distribution using **Student-t likelihood** for improved financial risk estimation.
- Presented findings at the **5th International Conference on Smart Systems (ICSS-2025)**.

Education

Central University of Rajasthan 2024–2026

M.Sc. Computer Science (Big Data Analytics)
CGPA: 8.0/10

- Coursework: Distributed Data Processing, Statistical Inference, Optimization Techniques

University of Delhi

2020–2023

B.Sc. Computer Science

CGPA: 8.56/10

- Coursework: Data Structures, Object-Oriented Programming, Database Systems

Achievements

- **2nd Place – Analytix'24 Data Analytics Competition**, Central University of Rajasthan.
- **Conference Presentation:** Research on Volatility Modeling presented at ICSS 2025.